## 9.10.5 Wrap-Up: Ticket Out the Door

The function $S(x) = -0.5x^2 + 50x - 8$, where $x$ is time in seconds (s), models the speed, $S$, in kilometers per hour (km/h) of Felix Baumgartner during the first $90$ seconds of his record-breaking free fall.

1. Use the function, $S(x)$, to determine when Felix reached the greatest speed.

2. Mathematical models use data to determine an equation that closely fits but may not be exact. According to the actual data collected during Felix's fall, the greatest speed he reached was $1355$ km/h, which occurred at $50$ seconds. How well did the function $S(x)$ match the real data?

# Unit 9, Lesson 11: Writing and Solving Quadratic Equations

### Benchmarks

MA.912.AR.3.1 Given a mathematical or real-world context, write and solve one-variable quadratic equations over the real number system.

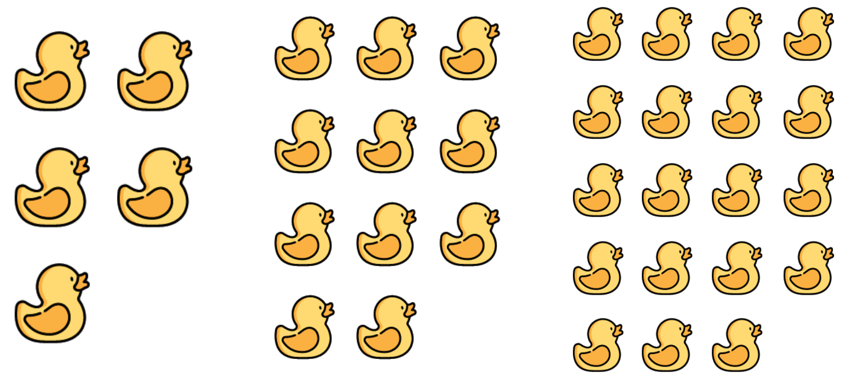
MA.912.AR.9.6 Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.

### Learning Target

☐ I can write and solve one-variable quadratic equations.

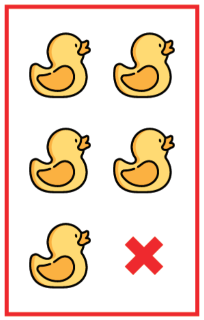
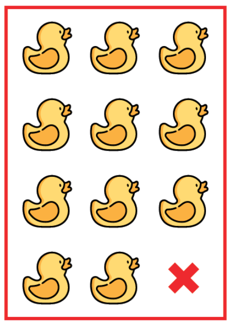
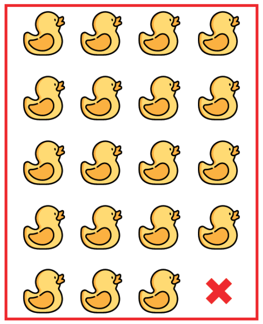
## 9.11.1 Warm-Up: Rubber Duck Quadratics

A pattern made of rubber ducks is shown.



1. Describe the pattern.

2. To describe the pattern, Emily made drawings and created an equation. Her drawings are shown below. Her equation is $D = (N + 1)(N + 2) - 1$.

| $N = 1$ | $N = 2$ | $N = 3$ |
|---|---|---|
|  |  |  |

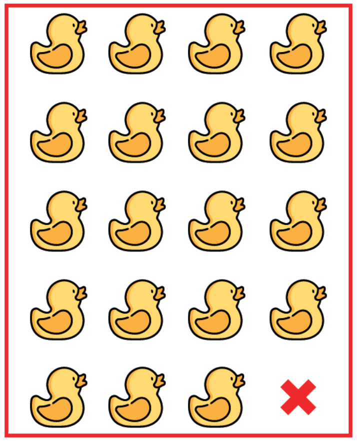
Use Emily's drawing for $N = 3$ to show how each expression from the equation $D = (N + 1)(N + 2) - 1$ is illustrated in her drawing.

**Expressions**

$(N + 1)$

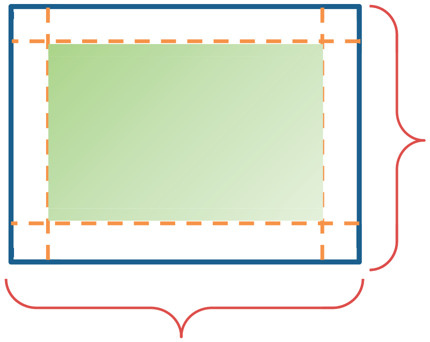
$(N + 2)$

$-1$



## 9.11.2 Guided Instruction: Writing and Solving Quadratic Equations

1. A company is experimenting with the size of a gift box without a top that can be made from a $12$ inch (in.) by $24$ in. piece of cardboard. The company wants to know if the height, $x$, of the box is changed, what the area of the bottom of the box will be.



   a. What is the width of the bottom of the box?

   b. What is the length of the bottom of the box?

   c. Discuss with your partner the constraints on the value of $x$, the height of the box.

   d. Write the function $A$ that can be used to determine the area, in square inches, in.$^2$, of the bottom of the box as a function of the height of the box, $x$.

   e. What is the height of the box if the area of the bottom of the box is $189$ in.$^2$?

   f. Discuss with your partner whether the solutions are reasonable for the given context.

2. A mathematician for a music company tracked the sales, in tens of millions of dollars, of music cassette tapes, $y$, for $x$ years since 1980 for the years 1980 to 2000. A table of values of some of the values is shown.

| $x$ | 0 | 1 | 3 | 6 | 9 | 10 | 13 | 17 | 19 |
|---|---|---|---|---|---|---|---|---|---|
| $y$ | 83 | 134 | 218 | 299 | 326 | 323 | 278 | 134 | 26 |

a. In what year did the sales of music cassette tapes reach its highest value?

b. Work with your partner to explain how the table can be used to confirm when the sales reached its highest value.

c. Write an equation that models the context.

d. In what year(s) were the sales 185 tens of millions of dollars?

3. Mangrove forests are important habitats for fish. An ichthyologist finds that some fish use mangrove forests only during high tide. The ichthyologist label fish that only use mangrove forests during high tide as "high-tide users." The ichthyologist gives values between 0 and 6.25 to represent the frequency of occurrence high-tide users are found in the mangrove forests. When the tide has been rising for 6.1 hours, the frequency of occurrence reaches a maximum of 6.25. The frequency of occurrence is 5 when the high tide has been rising for 3.6 hours. The frequency of occurrence reaches 5 again after 8.6 hours.

a. Define the variables.

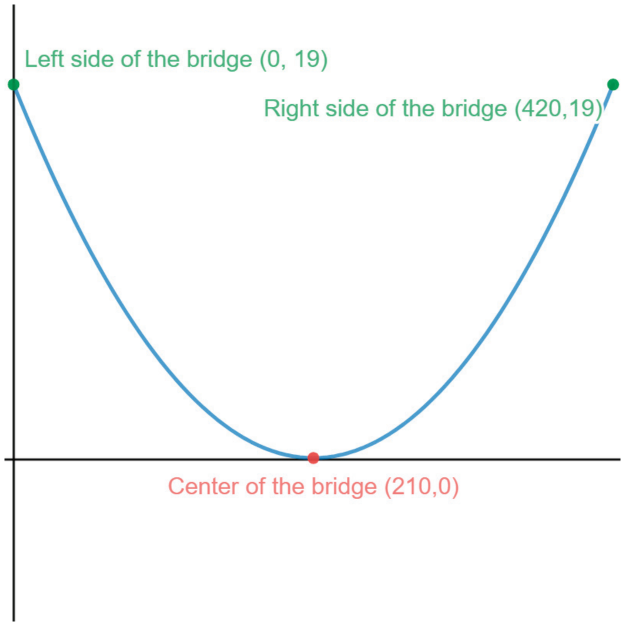
b. Write an equation, in vertex form, that models the context.

c. Determine when the frequency of occurrence will be 3.75.

## 9.11.3 Guided Practice: Writing and Solving Quadratic Equations

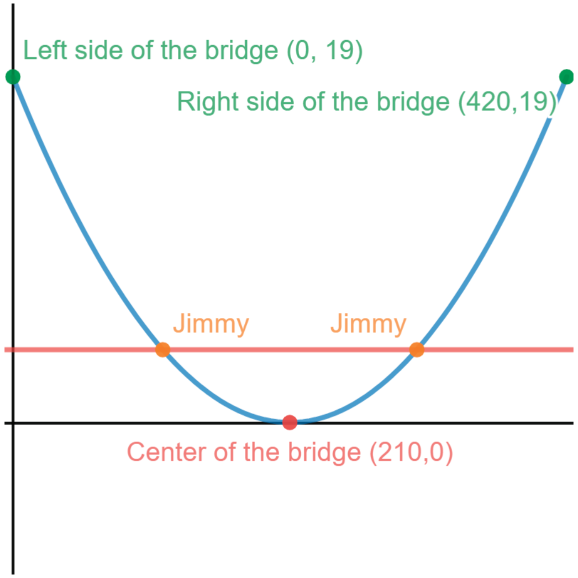
The Hal Adams bridge is a suspension bridge that crosses the Suwanee River near Mayo, Florida. The suspension wires are linked into and form a parabola, as shown in the photo of the bridge.


The lowest part of the parabola is located at the midpoint of the main span of the bridge, which is $420$ feet long. The highest points of the suspension are $19$ feet above the lowest point. The graph illustrates the parabola.




1. Write an equation that models the context. Leave the $a$-value as an unsimplified fraction.

2. Jimmy is $4$ feet tall. Where on the bridge will Jimmy be just as tall as the suspension cables?

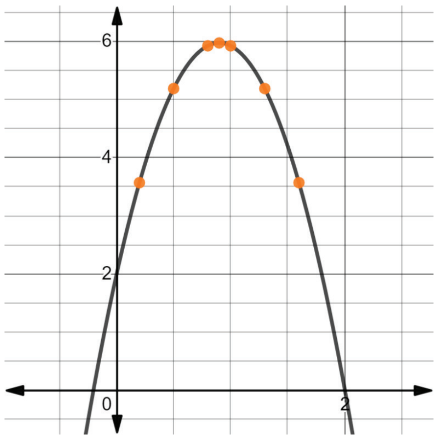


## 9.11.4 Your Turn!

Matias launches a water balloon from a deck that is $2$ feet off the ground. The time, in seconds (sec.), from the launch and the height, in meters (m), of the water balloon is given in the table.

| $t$ (sec.) | 0.2 | 0.5 | 0.8 | 0.9 | 1 | 1.3 | 1.6 |
|---|---|---|---|---|---|---|---|
| $h$ (m) | 3.568 | 5.185 | 5.92 | 5.969 | 5.92 | 5.185 | 3.568 |

A graph of the height of the water balloon is shown.




1. Write the function, $h(t)$, that models the height, $h$, in meters as a function of time, $t$, in seconds.

2. Determine when the water balloon will hit the ground.

## 9.11.5 Wrap-Up: Reflection

During the unit, you learned about writing and solving quadratics. Draw an **X** on each line to indicate your current level of understanding related to each concept.

1. Given the $x$-intercepts and another point on the graph of a quadratic function, write the equation for the function.

| I need help. I can't get started. | I'm getting there, but I need more practice. | I understand this and I feel confident. |
|---|---|---|

2. Write a quadratic function to represent the relationship between two quantities from a graph, a written description, or a table of values within a mathematical or real-world context.

| I need help. I can't get started. | I'm getting there, but I need more practice. | I understand this and I feel confident. |
|---|---|---|

3. Given a mathematical or real-world context, write and solve one-variable quadratic equations over the real number system.

| I need help. I can't get started. | I'm getting there, but I need more practice. | I understand this and I feel confident. |
|---|---|---|

4. Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.

| I need help. I can't get started. | I'm getting there, but I need more practice. | I understand this and I feel confident. |
|---|---|---|